Disturbance Observer

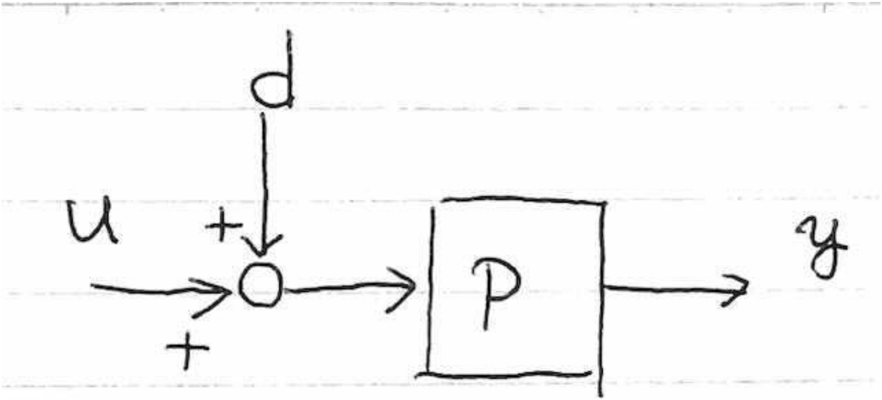
Electric, Electronic and System Engineering

without Disturbance Observer

Transfer Function

P: generalized plant
d: disturbance
u: input
y: output

The output y includes the influence of disturbance d.



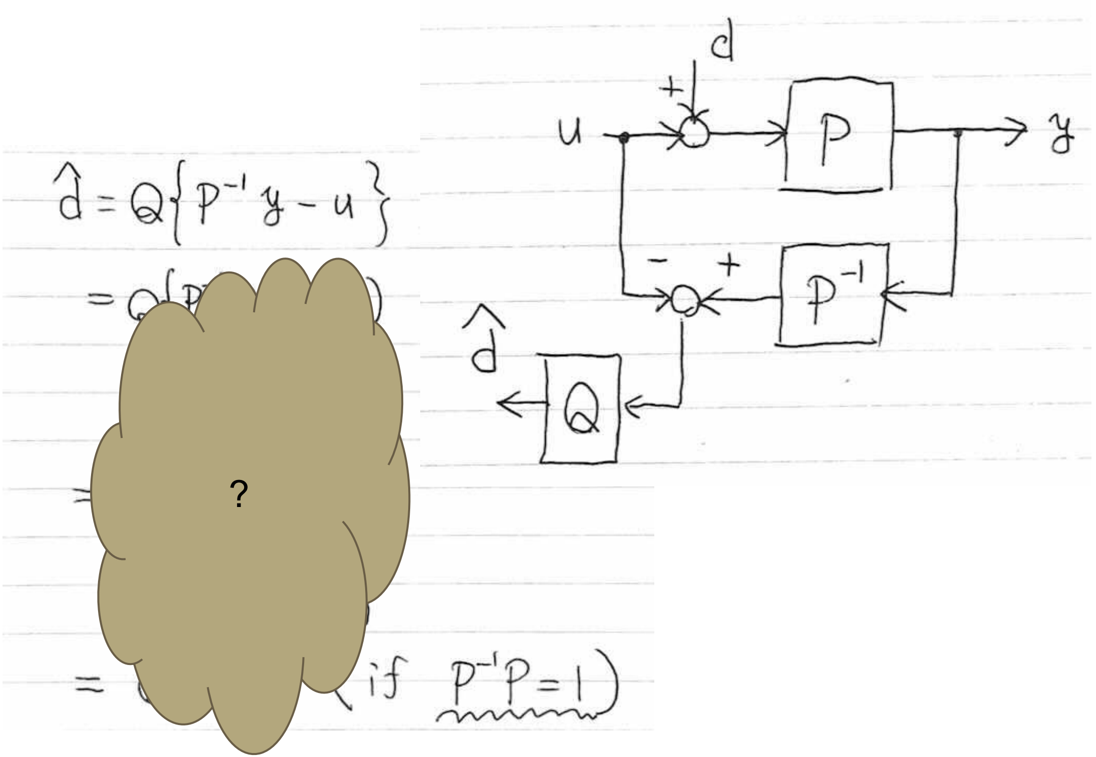
$$y = P(u+d)$$
$$= Pu + Pd$$

Disturbance Estimation

Computing the Inverse Model of Plant

P-1: Inverse Model of Plant P
d^ : Estimated Value of disturbance d
Q: Filter (Low Pass Filter is usual)

We can estimate the value of d using computation with P-1 & u & Q.



Disturbance Rejection

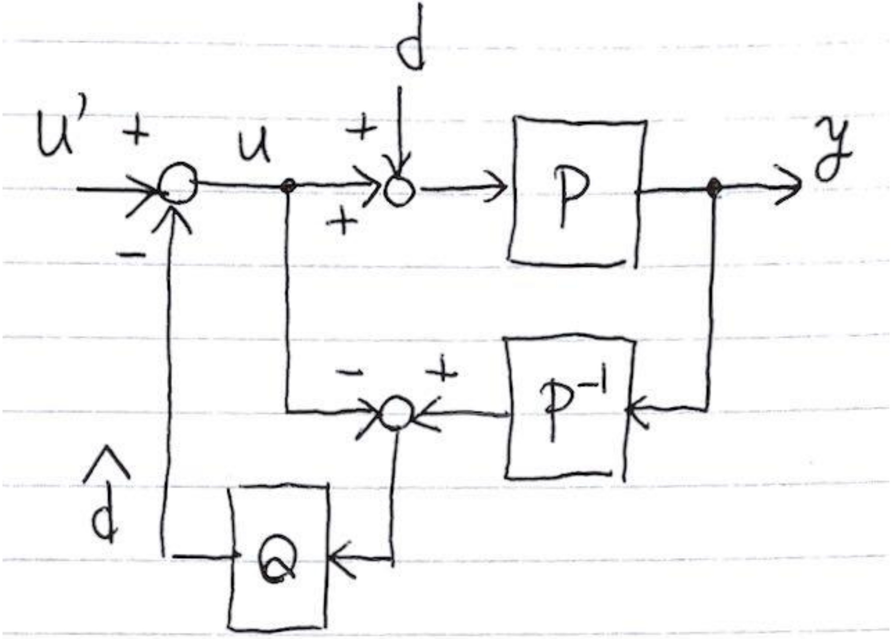
Introduce New Input u'

Then we can use d^ to reject the effect of disturbance d.

u' : input (≠u)

$$u = u' - \hat{d}$$

$$y = Pu + Pd$$



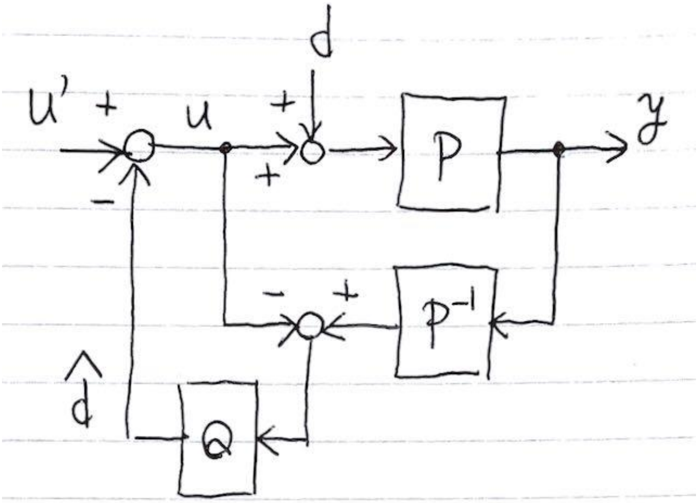
Express output y using input u' & disturbance d

$$\hat{d} = Q(P^{-1}y - u) = Q(P^{-1}y - u' + \hat{d})$$

$$(1-Q)\hat{d} = \ldots$$

?

$$\{(1-Q)+QPP^{-1}\}\,y = P(u' + (1-Q)d)$$



Ideal Case (PP-1 = 1)

PP-1 = 1

We can simplify the equation using PP-1 =1.

We got the equation output y using input u' & disturbance d.

If filter Q=1, we can get the output y without the influence of disturbance d.

$$PP^{-1} = 1,$$

$$\{(1-Q)+Q\}y = P\{u' + (1-Q)d\}$$

$$y = Pu' + (1-Q)Pd$$

if $Q \doteq 1$ $\quad y = Pu'$

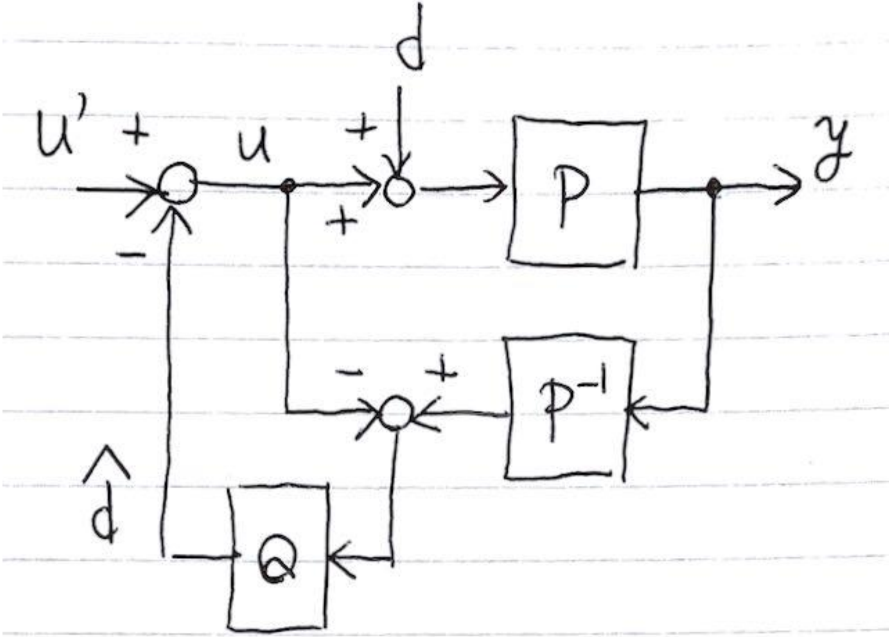
Filter Q?

Without filter Q (or Q=1), computation of u will be impossible.

if y=0, Q=1

u=u'-u,   u=(1/2)u' ????

In general case, low pass filter is used for Q.



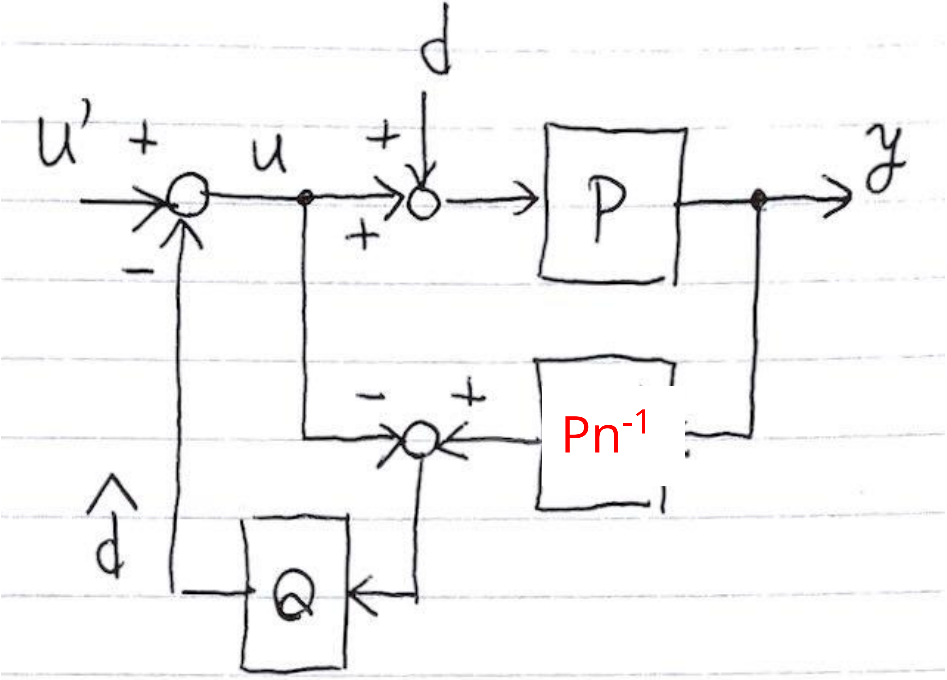
non Ideal case (PP-1 $\neq$ 1)

Introducing nominal plant Pn

Pn: nominal Plant

We can use the inverse model of nominal plant for computation of disturbance.

In this case, d^ includes the model error and disturbance.



Disturbance Observer nominalize the Plant

if Q≒1, we can get output y without disturbance and based on nominalized plant.

That is, we can modify the plant characteristics.

(ex. Big motor will perform as small motor.)

: $PP^{-1} \doteqdot 1 \rightarrow P^{-1} \rightarrow P_n^{-1}$

P_n : nominal Plant

$$\{(1-Q)+QPP_n^{-1}\}y = P(u'+(1-Q)d)$$

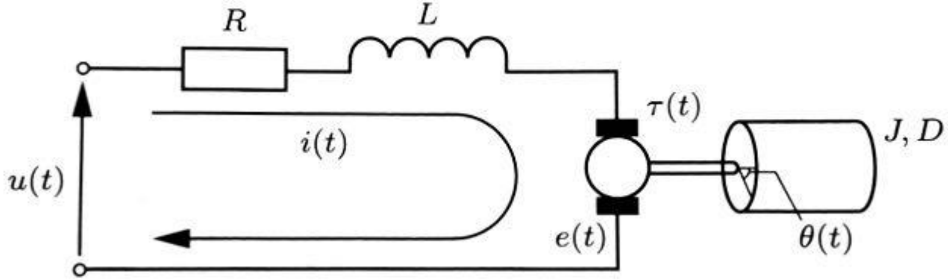
$$y = \frac{P(u'+(1-Q)d)}{\{(1-Q)+QPP_n^{-1}\}}$$

if $Q \doteqdot 1$

$$y = \frac{\cancel{P}(u'+(1-Q)d)}{\cancel{P}P_n^{-1}} = P_n(u'+0\cdot d)$$

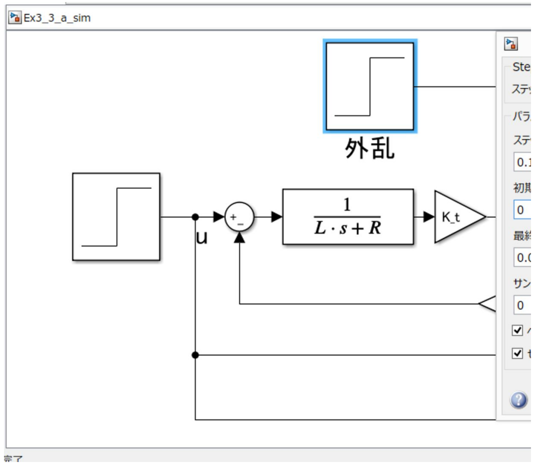
Implementation on Simulink

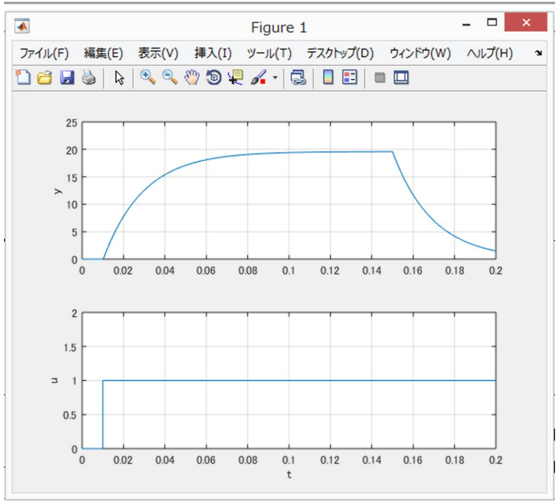
Motor Speed with Voltage Input



Check Slides for Motor Speed HomeWork.

The disturbance input drops the motor speed.

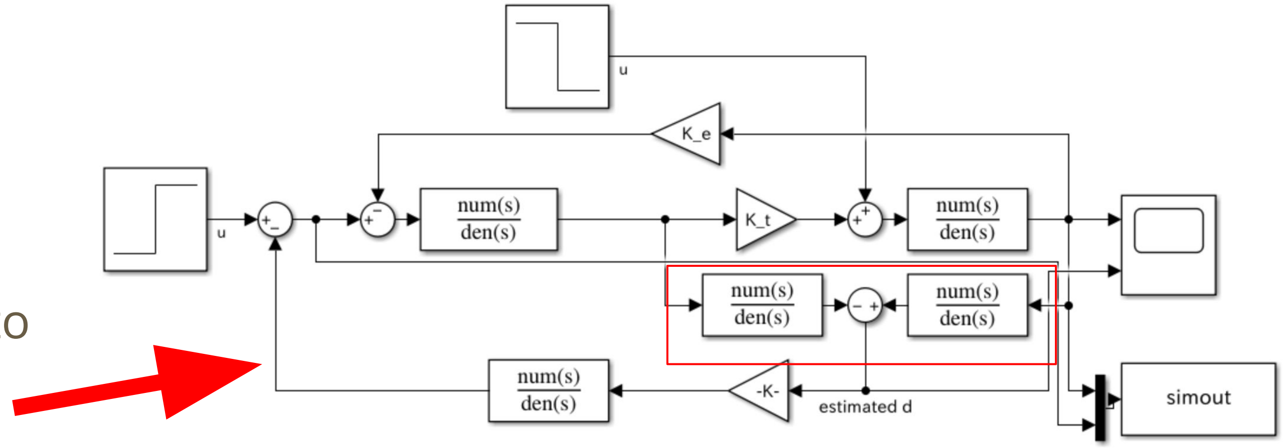




Motor Speed with DO

Lower part is disturbance observer with Low Pass Filter.

Additional voltage command is imputed to reject the effect of disturbance.



Control the simulation using .m file

Td: Time constant of low pass filter
→affects the estimation speed of disturbance observer

Endtime is changed to confirm the disturbance rejection using DO.

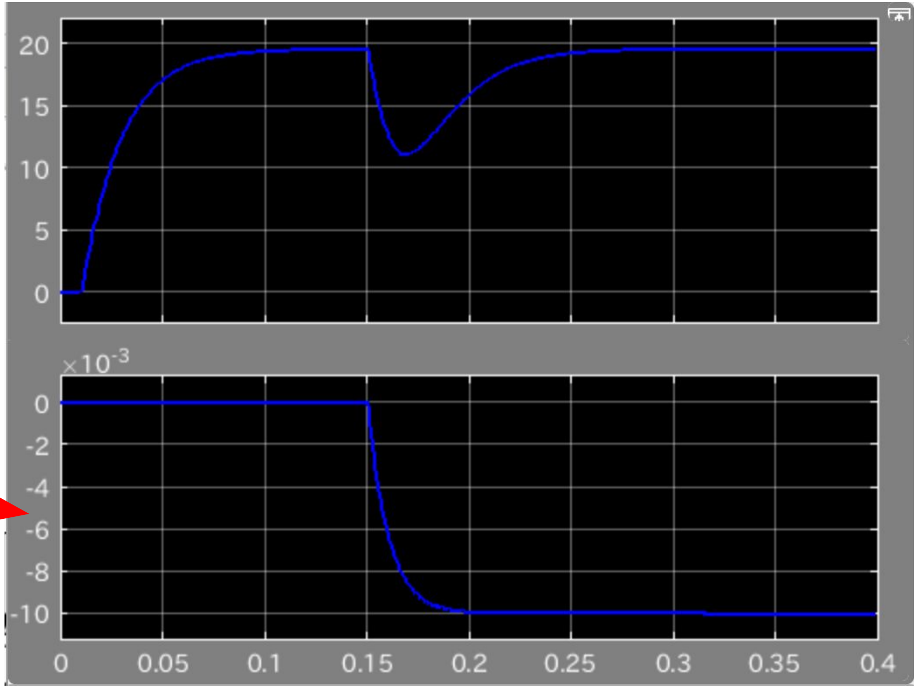
```
clear      % ワークスペースからすべての変数を消去
close all  % すべてのFigureを消去
clc        % コマンド ウィンドウのクリア

%システムの物理パラメータ
u = 1.0;    %入力電圧[V]
R = 5.0;    %電気子抵抗[Ω]
L = 1e-3;   %インダクタンス[H]
K_e = 5e-2; %逆起電力定数[Vs/rad]
K_t = 5e-2; %トルク定数[Nm/A]
D = 1e-5;   %粘性摩擦係数[Nmrad/s]
J = 1e-5;   %慣性モーメント[kgm^2]
Td = 0.01; %外乱オブザーバ時定数[s]


%Simulinkの実行
Endtime = 0.4;          %シミュレーション実行時間
filename = ['Ex3_d_sim']; %ファイル名（拡張子なし）
open(filename);         %Simulinkファイルを開く
sim(filename);          %Simulinkの実行
```

DO estimate the disturbance input

Motor speed drops due to the disturbance input at t=0.15s.

However, the speed will become the desired value within 0.1s.

DO estimates the disturbance input with 1st order time delay.



We need additional voltage to compensate d

Actually, we need additional voltage input to compensate the influence of disturbance.

In steady state, we need additional +1V for compensation.